

Karunadu Technologies Private Limited

Internship Project



Project Title

Crop Recommendation System using Machine Learning & Django



Introduction

In modern agriculture, selecting the right crop based on soil and environmental conditions is essential for maximizing yield and minimizing losses. Precision agriculture uses data-driven techniques to support farmers in making informed decisions.

In this assignment, students will develop a **Machine Learning-based web application** to recommend the most suitable crop based on soil nutrients and climatic conditions.



Dataset

The dataset contains various attributes such as:

- Nitrogen (N) – Soil nitrogen content (mg/kg)
- Phosphorus (P) – Soil phosphorus content (mg/kg)
- Potassium (K) – Soil potassium content (mg/kg)
- Temperature (°C)
- Humidity (%)
- pH – Soil pH value
- Rainfall (mm)

Interns will build and compare multiple classification models to predict the output label.



Objective

- Apply Machine Learning classification algorithms
- Understand data preprocessing and feature engineering
- Build a web application using Django
- Compare performance of different algorithms

Backend

Python
Django
Framework
Scikit-learn

Frontend

HTML
CSS / Bootstrap
Django Templates

Expected Output

- User login system
- Data input form
- Disease prediction output
- Accuracy comparison of models



Tasks

Interns must implement **at least 4 classification algorithms** from the list below:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Naive Bayes



Tasks

Web Application (Django)

Interns must build a **full-stack web application** using Django.

Login Module

- User Registration
- Login Authentication
- Session Management

Input Form Page

Build Django UI with input form of all Inputs

Prediction Module

Add a **“Predict” button**

On submission:

- Load trained ML models
- Predict output label
- Display result on UI

Results Display

- Predicted Output
- Accuracy of each algorithm
- Best performing algorithm



Important Questions

- What is classification in Machine Learning? How is it different from regression?
- Explain the importance of the output in this dataset.
- What is feature selection? Why is it important in this project?
- Explain the difference between supervised and unsupervised learning.
- What is overfitting and underfitting? How can it be avoided?
- What is the role of training and testing datasets?
- Explain how Logistic Regression works for classification.
- How does a Decision Tree decide splits?
- What is the advantage of Random Forest over Decision Tree?
- Explain how KNN works and its limitations.
- What is the concept behind SVM (Support Vector Machine)?
- What is a Confusion Matrix? Interpret it for your model.
- What preprocessing steps did you apply to the dataset?
- How did you handle missing or inconsistent data?
- Explain the flow of your Django application.
- How does authentication (login system) work in your project?
- What is the role of views, templates, and models in Django?
- How is the prediction triggered from the frontend?



Submission Requirements

Interns must submit

- Internship Report
- Source Code (Django + ML models)
- Dataset (if modified)
- Project Report (PDF) including:
 - Problem Statement
 - Methodology
 - Algorithms Used
 - Results & Comparison
 - Screenshots



Contact US



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**Thank You Interns for being a
part of Internship Journey**

**Contact Karunadu Technologies Private Limited
For Internship**